EIA GDP Indirect Impact Analysis

Navit Dori

March 20, 2025

1 Basic Leontieff equations

$$\begin{array}{ll} x & \text{Output} \\ II & \text{Intermediate input, also called IO matrix} \\ f & \text{Final demand} \\ T & \text{Technical coefficient matrix, sometimes called A} \end{array} \tag{1}$$

x is the total output vector, which is the total output productive sectors produce,

$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

II is the intermediate input matrix, or the IO matrix, which is what productive sectors buy and sell from other productive sectors,

Each element in the intermediate input matrix II is devided by the total output of the column (total output of the buying sector) to get the technical coeeficient:

$$T_{ij} = \frac{II_{ij}}{x_j} \tag{2}$$

This is an element-by-element multiplication, not a matrix multiplication. It is equivalent to

$$II_{ij} = T_{ij} x_j. \tag{3}$$

The fundamental equation states that the total output is a sum of the intermediate input and the final demand,

$$\begin{aligned} x_i &= II_i + f_i \\ &= \sum_k T_{ik} x_k + f_i, \end{aligned} \tag{4}$$

where $II_i = \sum_k T_{ik} x_k$ is what sector i sells to the other sectors, a sum of the row correxponding to sector i. f_i is what sector i sells to households.

In matrix form the equation becomes

$$\vec{x} = T\vec{x} + f \tag{5}$$

leading to

$$\vec{x} = (I - T)^{-1} f. \tag{6}$$

We usually refer to $(I - T)^{-1}$ as the Leontieff matrix, L.

2 Value Added

GDP is the same as value added

$$\text{Value Added} = \text{Output} - \text{Intermediate Input} \tag{7}$$

$$\text{Value Added} = \text{Wages} + \text{Taxes} - \text{Subsidies} + \text{Rent} + \text{Capital} + \text{Profit} \tag{8}$$

3 GDP Direct Impact

$$\begin{aligned} \text{Direct Impact} = \text{Wages} + \text{Taxes (of production)} - \text{Subsidies (of production)} \\ + \text{Operating Surplus} + \text{Capital} + \text{Profit}, \end{aligned} \tag{9}$$

or

$$\text{GDP}_{\text{Direct Impact}} = \frac{VA}{Output} = \frac{GDP}{Output} \tag{10}$$

$$VA = GDP \tag{11}$$

The above formula has the advantage that for any new total output vector $\vec{x'}$ the new GDP is $GDP_{\text{Direct Impact}}\vec{x'}$, which is to say that for sector i the new GDP is $GDP_{\text{Direct Impact},i}\vec{x'}_i$

4 GDP indirect impact - our EIA model version

Here I follow the power point file *EIA Modelling Workshop-Day 1_Dec 23 2019.pptx*. Reetika explained over the phone that the vector E' is based on $II' = Outcome' - ValueAdded'$. All variable denoted by $'$ refer to variables after the shock. In the example given in the file (slide 21),

$$II'_a = Output'_a - ValueAdded'_a = 5 - 3 = 2. \tag{12}$$

Value added is refered to as **GDP direct** in the file.

The next step is

$$E'_a = T_{i1}(output'_a - \mathrm{GDP}'_a) = T_{i1} \cdot \mathrm{II}'_a \tag{13}$$

where T_{i1} is the technical coefficients of the first column of T, refering to sector a. We could write $T_{i1} = T_{ia}$. The first column of T is $[0.00, 0.17, 0.33]$ and so we have

$$E' = [0.00, 0.17, 0.33] \cdot 2 = [0.00, 0.34, 0.66] \tag{14}$$

The next step is to multiply E' by the inverse Leontieff matrix L as if it were a final demand variable. The equation given in slide 21 is

$$Y' = (I - A)^{-1}E' \tag{15}$$

where in the file's notation Y' is the total output (here x') and A is the technical coefficient (here T).

Y' is thus

$$Y' = \begin{pmatrix} 1.154 & 0.385 & 0.269 \\ 0.322 & 1.275 & 0.319 \\ 0.503 & 0.385 & 1.325 \end{pmatrix} \begin{pmatrix} 0.00 \\ 0.34 \\ 0.66 \end{pmatrix} = \begin{pmatrix} 0.31 \\ 0.64 \\ 1.01 \end{pmatrix} \tag{16}$$

The file continues with

$$\mathrm{GDP}typeI = Y' \cdot \mathrm{GDP}_{\mathrm{ratio}} \tag{17}$$

where $\mathrm{GDP}_{\mathrm{ratio}}$ is calulated in slide 18 to be

$$\text{GDP to output ratio of sector j} = \frac{\mathrm{GDP}_j}{x_j} \tag{18}$$

and the result is

$$\text{GDP to output ratio} = [0.50, 0.47, 0.55] \tag{19}$$

However, slide 21 uses a different vector

$$\text{GDP}ratio = [0.44, 0.37, 0.50]$$

The final step is to carry out the multiplication of eq. 17 and sum over the result

$$\begin{aligned} \text{GDP}typeI_a &= Y' \cdot \text{GDP}ratio \\ &= [0.31, 0.64, 1.01] \cdot [0.44, 0.37, 0.50] \\ &= [0.14, 0.24, 0.50] \end{aligned} \tag{20}$$

$$\text{GDP}indirect_a = \sum_i Y'_i \cdot \text{GDP}ratio_i = 0.14 + 0.24 + 0.50 = 0.88 \tag{21}$$